Liam Fowl

EDUCATION

PhD in Mathematics Sept 2016 - May 2022
University of Maryland, USA
Advised by Professor Tom Goldstein (CS) and Professor Wojtek Czaja (Math).
GPA: 4.0/4.0

BS in Mathematics Sept 2012 - May 2016
University of Maryland, USA
High honors in Mathematics, with a minor in Physics.
GPA: 3.9/4.0

EXPERIENCE

Postdoctoral Researcher - Princeton University

Princeton Language and Intelligence (advised by Prof. Sanjeev Arora) August 2025 - Present

- LLM reasoning for both formal (Lean) and informal Mathematics.
- Post-training methods (on-policy self-distillation, reinforcement learning (RL)).

Senior Research Scientist - Google DeepMind

Gemini - tool use September 2023 - July 2025

- Led effort to build preference data for Gemini tool use for several extensions.
- Authored production RFCs (including as lead author) and contributed to Gemini's production SFT data mixture.

Speech Research Team (federated) June 2022 - September 2023

- Led an i18n federated training routine (custom locale teachers and recipes); defensive patent publication on federated hotword label prompting (primary author, dpub# 6232).

PUBLICATIONS & PREPRINTS

** denotes equal contribution; † denotes equal senior authorship.*

[1] Dang, X., Agarwal, R., Porto, R., Goyal, A., **Fowl*, L.**, & Arora, S. *Escaping the Cognitive Well: Efficient Competition Math with Off-the-Shelf Models.* AI for Math Workshop, **ICML** 2026. *Under review at NeurIPS 2026.*

[2] Chung, J., Cai, Z., Li, Z., Yin, Q., Agarwal, R., Park, S., Porto, R., Ri, N., et al., **Fowl*†, L.**, & Arora, S. *Goedel-Architect: Streamlining Formal Theorem Proving with Blueprint Generation and Refinement.* AI for Math Workshop, **ICML** 2026.

[3] He, Y., Kaur, S., Bhaskar, A., Yang, Y., Liu, J., Ri, N., **Fowl, L.**, Panigrahi, A., Chen, D., & Arora, S. *Self-Distillation Zero: Self-Revision Turns Binary Rewards into Dense Supervision.* ICML 2026 RLxF Workshop; Spotlight, ICML 2026 AI for Math Workshop.

[4] Kaur, S., Ri, N., He, Y., **Fowl, L.**, & Arora, S. *Rethinking On-Policy Self-Distillation for Thinking Models.* AI for Math Workshop, **ICML** 2026; *under review at NeurIPS 2026.*

[5] Lazic, N., **Fowl, L.**, György, A., & Szepesvári, C. *To See the Unseen: On the Generalization Ability of Transformers in Symbolic Reasoning.* Accepted for publication at Conference on Language Modeling, **CoLM** 2026.

[6] Kalra, N., Ri, N., Bukhari, Z., Li, A., Lotfi, S., **Fowl, L.**, & Goldblum, M. *Measuring the Limits of Continual Learning for LLMs.* Workshop on Compositional Learning, **ICML** 2026.

[7] **Fowl*, L.**, Geiping*, J., Reich, S., Wen, Y., Goldblum, M., & Goldstein, T. *Decepticons: Corrupted Transformers Breach Privacy in Federated Learning for Language Models.* International Conference on Learning Representations **(ICLR)** 2023.

[8] **Fowl*, L.**, Geiping*, J., Czaja, W., Goldblum, M., & Goldstein, T. *Robbing the FED: Directly Obtaining Private Data in Federated Learning with Modified Models.* International Conference on Learning Representations **(ICLR)** 2022.

[9] **Fowl*, L.**, Goldblum*, M., Chiang, P., Geiping, J., Czaja, W., & Goldstein, T. *Adversarial Examples Make Strong Poisons!* Advances in Neural Information Processing Systems **(NeurIPS)**, 2021.

[10] **Fowl, L.**, Partridge, K., Chen, N., Moreno, I.L. *Improved Labeling for Federated Keyword Spotting Training via Label Prompting.* Under review.

[11] Souri, H*, **Fowl*, L.**, Chellappa, R., Goldblum, M., & Goldstein, T. *Sleeper Agent: Scalable Hidden Trigger Backdoors for Neural Networks Trained from Scratch.* Advances in Neural Information Processing Systems **(NeurIPS)**, 2022.

[12] Somepalli, G., **Fowl, L.**, Bansal, A., Yeh-Chiang, P., Dar, Y., Baraniuk, R., Goldblum, M., Goldstein, T. *Can You Learn the Same Model Twice? Investigating Reproducibility and Double Descent from the Decision Boundary Perspective.* Conference on Computer Vision and Pattern Recognition **(CVPR)**, 2022. *Oral.*

[13] Geiping*, J., **Fowl*, L.**, Huang, W. R., Czaja, W., Taylor, G., Moeller, M., & Goldstein, T. *Witches' Brew: Industrial Scale Data Poisoning via Gradient Matching.* International Conference on Learning Representations **(ICLR)**, 2021.

[14] Wen*, Y., Geiping*, J., **Fowl*, L.**, Goldblum, M., & Goldstein, T. *Fishing for User Data in Large-Batch Federated Learning via Gradient Magnification.* International Conference on Machine Learning **(ICML)** 2022.

[15] Chu, H, Geiping, J., **Fowl, L.**, Goldblum, M., & Goldstein, T. *Panning for Gold in Federated Learning: Targeted Text Extraction under Arbitrarily Large-Scale Aggregation.* International Conference on Learning Representations **(ICLR)** 2023.

[16] Goldblum*, M., **Fowl*, L**., & Goldstein, T. *Adversarially robust few-shot learning: A meta-learning approach.* Advances in Neural Information Processing Systems **(NeurIPS)**, 2020.

[17] Goldblum*, M., **Fowl*, L.**, Feizi, S., & Goldstein, T. *Adversarially robust distillation.* In Proceedings of the **AAAI** Conference on Artificial Intelligence 2020.

[18] Goldblum, M., Reich*, S., **Fowl*, L.**, Ni*, R., Cherepanova*, V., & Goldstein, T. *Unraveling meta-learning: Understanding feature representations for few-shot tasks.* International Conference on Machine Learning **(ICML)** 2020.

[19] Huang*, W. R., Geiping*, J., **Fowl, L.**, Taylor, G., & Goldstein, T. (2020). *Metapoison: Practical general-purpose clean-label data poisoning.* Advances in Neural Information Processing Systems **(NeurIPS)**, 2021.

[20] Geiping, J., **Fowl, L.**, Somepalli, G., Goldblum, M., Moeller, M., & Goldstein, T. (2020). *What Doesn't Kill You Makes You Robust(er): Adversarial Training against Poisons and Backdoors.* https://arxiv.org/abs/2102.13624. Under review.

[21] Borgnia*, E., Cherepanova*, V., **Fowl*, L.**, Ghiasi*, A., Geiping*, J., Goldblum*, M., ... & Gupta*, A. *Strong Data Augmentation Sanitizes Poisoning and Backdoor Attacks Without an Accuracy Tradeoff.* In IEEE International Conference on Acoustics, Speech and Signal Processing **(ICASSP)** 2021.

[22] Abdelkader*, A., Curry*, M. J., **Fowl*, L.**, Goldstein*, T., Schwarzschild*, A., Shu*, M., ... & Zhu*, C. *Headless Horseman: Adversarial Attacks on Transfer Learning Models.* In IEEE International Conference on Acoustics, Speech and Signal Processing **(ICASSP)** *(Spotlight)* 2020.

[23] Labrador, B., Zhao, G., Moreno, I.L., Scarpati, A., **Fowl, L.**, Wang, Q., *Exploring Sequence-to-Sequence Transformer-Transducer Models for Keyword Spotting.* In IEEE International Conference on Acoustics, Speech and Signal Processing **(ICASSP)** 2023.

[24] **Fowl*, L.**, Goldblum*, M., Gupta, A., Sharaf, A., & Goldstein, T. (2020). *Random Network Distillation as a Diversity Metric for Both Image and Text Generation.* **NeurIPS** 2020 Workshop on Dataset Curation.

[25] **Fowl*, L.**, Chiang*, P., Goldblum*, M., Geiping, J., Bansal, A., Czaja, W., & Goldstein, T. *Preventing Unauthorized Use of Proprietary Data: Poisoning for Secure Dataset Release.* In **NeurIPS** 2020 Workshop on Dataset Curation.

[26] Huang, W. R., Emam, Z., Goldblum, M., **Fowl, L.**, Terry, J. K., Huang, F., & Goldstein, T. *Understanding generalization through visualizations.* arXiv:1906.03291.

[27] Peri, N., Gupta, N., Huang, W. R., **Fowl, L.**, Zhu, C., Feizi, S., ... & Dickerson, J. P. *Deep k-nn defense against clean-label data poisoning attacks.* In European Conference on Computer Vision **(ECCV)**AROW workshop.

MANUSCRIPTS IN PREPARATION

Fowl, L., Arora, S., & Razin, N. *What Can Your Policy See? Investigating Visibility Effects for RL Rewards.*

Fowl, L., Ortega, T., Khaitan, A., Kontorovich, A., & Arora, S. *Goedel-Architect v2.*

Strauss, H., **Fowl, L.**, & Razin, N. *Learning Dynamics of On-Policy Distillation.*

COMMUNITY INVOLVEMENT

Reviewer - ICLR: 2020, 2021, 2022, 2023, 2024, 2025 NeurIPS: 2020, 2021, 2022, 2023, 2024 CVPR: 2021

Organizer Deep Learning Research Interaction Team (RIT) at UMD — Sept 2018 - 2022

Organizer - Norbert Wiener Center Seminar — Sept 2018 - 2022

Volunteer - Girls Excelling in Math and Science (GEMS) — Sept 2016 - Jun 2018

Tutor - Northwestern High School (Prince George's County MD) — Sept 2015 - Jun 2018

Mentor - Directed Reading Program at UMD — Sept 2016 - Jun 2018

AWARDS

Wiley Dissertation Fellowship - Top university-wide dissertation fellowship.

Banneker-Key Scholarship - Most prestigious full scholarship given to top 1% of incoming students at UMD.

Aziz Mathematics Scholarship - Given once per year to the top undergraduate student in Mathematics.

Outstanding Senior Award - Given at the end of each year to the top graduating senior in Mathematics each year.

Award for Excellence in Teaching - Given to graduate students who have demonstrated excellent teaching abilities.